



GAME DESIGN RESEARCH LAB

From Playing to Building: Design Your Own Shooting & Survival Game

CS 4700 · Game Development | Cal Poly Pomona



Activity Overview

In this activity you will investigate two real Unity game templates, one is the first-person shooter and then you do research on a multiplayer survival game. Then you use what you discovered to pitch an original game concept. Work through each section, fill in every answer box, and be ready to share your ideas with the class on Thursday, April 16.

Estimated time: 75 minutes | **Team size:** 2–3 students |

Deliverable: Completed worksheet + 3-minute team pitch



PART 1 — Meet the Games (20 min)

Read, explore, and take notes on the two Unity templates you will be designing from.



Game 1: Sharp Shooter (Single-Player FPS)

A first-person shooter built in Unity using the Starter Assets Character Controller. The player navigates indoor environments, fights AI robot enemies and turrets, manages ammo, picks up weapons, and tries to survive waves of increasingly dangerous foes.



Shooting System

Hit scan & projectile weapons via Weapon scriptable objects; recoil, zoom, ammo counter



Enemy AI

Hovering robots, turrets, spawn gates, each with health, patrol, and attack logic



Health & Shield

Player Health component tracks HP; UI shows shield bar, health bar, and death state



Pickups

Ammo Pickup & Weapon Pickup scattered across levels; trigger-based collection



Environment

Modular indoor rooms with doors, corridors, and strategic cover



Controls

First Person Controller (WASD + mouse look) with sprint, jump

Game 2: Survival — Multiplayer Survival

A multiplayer survival engine by Mirror Networking. Players join a persistent world, fight zombies, manage vital stats (hunger, thirst, temperature), craft items, harvest resources, and build shelters, together or against each other.

Name your inspirational survival game: [Zombie Survival](#)

Multiplayer	Mirror Networking; server-authoritative; supports 100+ concurrent users
Zombie AI	loot drops, runtime spawning, all tunable in Inspector
Survival Stats	Health, Nutrition, Hydration, Temperature, Endurance, all drain over time
Crafting	Recipe-based crafting system; smelting furnace; item durability
Harvesting	Chop trees, gather materials, respawning resources in the world
Building	Place structures in the world; modular build system; SQLite persistence



PART 2 — Research Challenge (20 min)

Explore 3 popular FPS and survival games. Fill in the table as a team.

Research 3 existing FPS / Survival Adventure games (not the ones above). You can use YouTube, Wikipedia, Steam, or Itch.io. Try to include at least one multiplayer and one single-player title.

Game Title	Genre / Mode	Core Mechanic	Unique Feature	Platform
Unturned	Zombie Survival Sandbox	Scavenging for resources, crafting items, and building structures to withstand zombie threats and enemy players	Weapon customization and modding freedom	PC and consoles
Project Zomboid	Top-down Zombie Survival	Managing health, weapons, and other resources while exploring new environments and defending against zombies	In-depth crafting system + thoughtful map design	PC only
Rust	First-person Post Apocalypse Survival	Resource gathering, crafting, base building, and PvP combat	Sandbox base building Base raiding / PvP combat	PC and consoles

What common patterns did you notice across these games? (e.g., HUD elements, enemy types, progression systems)

Health bars, hunger/thirst bars
Inventory management
Armored zombies, fast zombies
Crafting systems
Base Building
Gun play

Which game impressed you the most and why? What ONE feature would you steal for your own game?

Base Building
Inventory Management

PART 3 — Compare the Templates (10 min)

Think critically about Sharp Shooter vs Survival as starting points for your own project.

Feature	Sharp Shooter (FPS)	Survival (Survival)
Player perspective	First-person only	First or third person
Multiplayer?	Single-player only	Built-in (Mirror)
Enemy behavior	Robots & turrets (AI scripts)	Zombies with aggro & loot
Resource system	Ammo & weapon pickups	Crafting, harvesting, building
Survival stats	Health + shield bar only	HP + hunger, thirst, temp, stamina
Code complexity	Low (~7 custom scripts)	Medium (Mirror networking required)
Best used for...		

If you had to start a game project today, which template would you choose and why? Consider scope, features, and your team's skills.

PART 4 — Design Your Game Concept (25 min)

Use your research to design an original game. Fill in EVERY field — your answers become your class pitch!

Your game can remix, combine, or expand either template — or take a completely new direction inspired by your research. Be creative!

 Game Title	
 Team Members	
 Genre	<i>FPS / Survival / Battle Royale / Hybrid / Other:</i> _____
 Multiplayer Mode	<i>Single-Player / Co-op / PvP / MMO-style / Async</i>
 Base Template	<i>Sharp Shooter / uSurvival / Both / Neither (start fresh)</i>

Setting & Story: Describe the world your game takes place in. What year is it? What happened? Why is the player here?

Core Game Loop: What does the player DO every minute? (Hint: Shoot → Loot → Craft → Survive?)

Game Mechanics Design Table

Choose at least 4 mechanics and describe how YOUR game implements them.

Mechanic	Will you include it?	How does YOUR version work?
Weapons / Combat	Yes / No / Modified	
Health & Damage	Yes / No / Modified	
Survival Stats (hunger/thirst)	Yes / No / Modified	
Enemy AI	Yes / No / Modified	
Crafting System	Yes / No / Modified	
Multiplayer Networking	Yes / No / Modified	
Building / Base Construction	Yes / No / Modified	
Inventory / Loot Drops	Yes / No / Modified	
Unique mechanic (YOUR idea!)		

What makes your game UNIQUE? What feature have you never seen in another game?

Describe your visual style and audio vibe. (e.g., dark post-apocalyptic + heavy metal? Colorful cartoon + chiptune?)

PART 5 — Pitch Prep Checklist (5 min)

Get ready for your 3-minute team pitch to the class!

Your pitch must cover all 5 points below. Check each one as you finish preparing it:

- ☐ Game title and one-sentence hook (e.g., "It's Minecraft meets Call of Duty in a haunted space station")
- ☐ The base Unity template(s) you chose and ONE key feature you kept
- ☐ ONE feature you heavily modified or replaced — and how
- ☐ Your game's unique mechanic (the thing no one else has)
- ☐ Who is your target audience and on what platform would you release?

Your 3-minute pitch outline (bullet points are fine — just capture the key ideas you'll say aloud):

 **Sharp Shooter** · github.com/YourCourse/sharp-shooter | **Survival** · assetstore.unity.com
(search uSURVIVAL)

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